MANUFACTURING QUALITY ANALYSIS — RTY IMPROVEMENT CASE STUDY

Duke Cricket Ball — Quality Analysis & Defect Reduction

Emmanuella Owa

Chemical Engineer | Process Improvement & Quality Professional

Identified Print Logo (Step 3) as the critical process outlier — DPU 0.16 vs process average 0.08 — and prioritised FMEA corrective actions targeting RPN 336 stitch failure mode to improve Rolled Throughput Yield from 66% to a target of 88%.

Analysis conducted as part of graduate coursework in Process Quality Engineering, Conestoga College — applied using Duke & Son Ltd manufacturing process data.

TOOLS & METHODS

SIPOC · Cause & Effect (C&E) Matrix · Pareto Analysis · FMEA (SEV / OCC / DET / RPN) · FTY · DPU · Rolled Throughput Yield (RTY)

OVERVIEW

Executive Summary

> Duke's cricket ball manufacturing process was producing a Rolled Throughput Yield (RTY) of just 66% across 5 production steps — meaning only 66 in every 100 balls made it through the full process without a defect or rework. This analysis applied three complementary tools — SIPOC, C&E Matrix, and FMEA — to identify the vital few inputs and highest-risk failure modes, then quantified the quality baseline using FTY, DPU, and RTY metrics to prioritise the highest-impact improvements.

Business Problem

Industry	Sporting Goods Manufacturing — Premium Cricket Balls (Duke & Son Ltd)
Product	Hand-crafted cricket balls — premium market; MCC regulation compliance required
Problem	Process RTY of 66% indicates significant hidden rework and defect cost. Root causes not formally identified — improvement efforts unfocused.
Scope	5-step process: Wind Cores → Assemble Leather → Print Logo → Final Shape → Final Inspection
Data	6,450 total units inspected across all 5 steps; 528 total defects recorded
Target	Improve RTY from 66% to 88%+ through FMEA-driven corrective actions on vital few steps

Objectives

- Build a SIPOC to define process boundaries, inputs, outputs, suppliers, and customers.
- Apply C&E Matrix to score each process step against 7 customer-critical quality requirements — identifying the vital few steps driving quality outcomes.
- Execute FMEA on the top 3 C&E steps — quantify RPN and define corrective actions for highest-risk failure modes.
- Calculate FTY, DPU, and RTY across all 5 steps to baseline current quality performance.

APPROACH

Process Analysis

Step 1 — SIPOC: Duke Cricket Ball Manufacturing

The SIPOC maps the end-to-end process from raw material order through to the final consumer, defining all 8 process stages, their inputs, outputs, suppliers, and customers. This established the scope boundary and confirmed which steps to include in the C&E Matrix scoring.

Supplier	Input	Process Step	Output	Customer
Duke's Factory	Required RM list	Place raw material order	Confirmed order	Materials supplier
Materials supplier (Scotland)	Confirmed order	Package & ship raw materials	Delivered raw materials	Duke's factory
Sewer	Leather, thread, tools	Cut & sew leather into cups	Cup-shaped leather	Sewer (next stage)
Shaper	Seamer machine + ball	Shape & form cricket ball	Shaped cricket ball	Tester
Tester	Scale, bounce tester, sizer	Weight, size & bounce test	Tested ball / rework	Stamper / Rework
Stamper/Miller/ Polisher	Heat sealer, lamp, grease	Stamp, mill, lamp & polish	Finished cricket ball	Final Tester
Final Tester (Founder)	Eyes, hands, tested ball	Visual & feel test	Approved or rework	Packager
Packager	Box, approved ball	Package finished ball	Packaged product	Final Consumer

APPROACH (continued)

Step 2 — Cause & Effect (C&E) Matrix

How to read this matrix: Each process step is scored against 7 customer quality requirements using a 0/1/3/9 correlation scale (9 = strong impact, 3 = moderate, 1 = weak, 0 = none). Each score is multiplied by the requirement weight (7–10) and summed to produce a weighted total. Higher totals = greater influence on customer quality. The top 3 steps by weighted total become the focus of FMEA.

Scoring Scale: 9 = Strong relationship · 3 = Moderate · 1 = Weak · 0 = None

Process Step	Correct Weight Wt=10	Correct Size Wt=10	Bounce Height Wt=9	Seam Strength Wt=9	Surface Finish Wt=8	Logo Quality Wt=7	MCC Co mpliance Wt=9	Weighted Total
Score: 9=Strong 3=Moderate 1=Weak 0=None								
Weight/Size/Bounce Test	9	9	9	3	1	—	9	295
Sew Cups (6×80 stitches)	3	1	3	9	3	1	9	237
Cut & Sew Leather Strips	3	3	3	9	3	—	9	233
Quality Raw Materials	3	3	3	9	9	1	3	228
Shape/Swing Final Test	9	9	9	1	1	—	3	218
Stamp/Mill/Lamp/Polish	1	1	1	1	9	3	1	144
Shape Ball	3	3	1	1	3	—	3	117
Package Ball	1	1	1	1	3	3	1	116

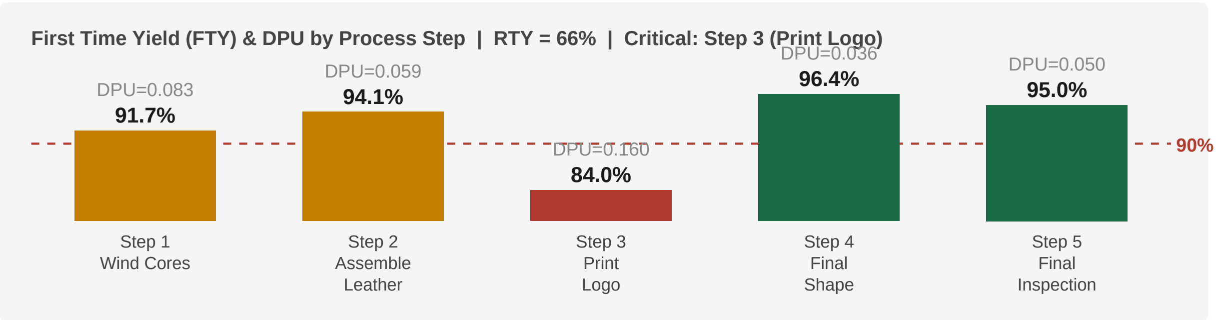
Vital few identified: Weight/Size/Bounce Test (295), Sew Cups 6×80 stitches (237), and Cut & Sew Leather Strips (233). These three steps score highest across the most heavily-weighted customer requirements — weight, size, bounce, seam strength, and MCC compliance — and become the inputs to the FMEA on the following page.

RESULTS

Step 3 — Quality Metrics: FTY, DPU & RTY

66%	0.08	84%	528
Rolled Throughput Yield	Overall Avg DPU	Worst FTY — Step 3	Total Defects (6,450 units)

FTY & DPU by Process Step



RTY Calculation

Step	Process	Units In	Defects	FTY	DPU	RTY Contribution
1	Wind Cores	1,500	125	91.7%	0.083	×0.917
2	Assemble Leather	1,500	89	94.1%	0.059	×0.941
3	Print Logo	1,500	240	84.0%	0.160	×0.840
4	Final Shape	1,200	43	96.4%	0.036	×0.964
5	Final Inspection	750	31	95.9%	0.041	×0.959
	TOTAL / RTY	**6,450**	**528**	—	**0.083 avg**	**0.917×0.941×0.840×0.964×0.959 = 66%**

Step 3 (Print Logo) is the critical outlier: FTY 84% and DPU 0.160 — both twice worse than the process average. It contributes the largest single drag on RTY. Yet Step 3 did not rank in the top C&E inputs — revealing it as a process execution problem (standardisation, tooling) rather than a design-level quality issue.

RESULTS (continued)

Step 4 — FMEA: Failure Mode & Effects Analysis

FMEA applied to the top 3 C&E steps. Each failure mode is rated on three dimensions: **Severity (SEV)** — impact on customer if failure occurs; **Occurrence (OCC)** — likelihood of failure occurring; **Detection (DET)** — likelihood of failure reaching the customer undetected. RPN = SEV × OCC × DET. Higher RPN = higher priority for corrective action.

Process Step	Failure Mode	SEV	OCC	DET	RPN	Priority	Rationale Summary
Sew Cups (6×80)	Incorrect stitch count or loose stitches	8	6	7	336	CRITICAL	SEV 8: seam failure ruins ball in play OCC 6: manual counting error common DET 7: hard to detect until seam fails
Bounce/Size Test	Ball fails bounce or size test	9	6	6	324	CRITICAL	SEV 9: direct MCC non-compliance OCC 6: subjective founder judgement DET 6: visual test only
Cut & Sew Strips	Leather cut to wrong dimensions	7	5	6	210	HIGH	SEV 7: affects seam + size OCC 5: manual cutting variation DET 6: not always caught at sewing
Quality Raw Materials	Substandard leather from supplier	8	3	8	192	HIGH	SEV 8: poor leather = surface defects OCC 3: supplier fairly reliable DET 8: no incoming inspection
Bounce/Size Test	Ball fails weight test	9	5	4	180	HIGH	SEV 9: MCC non-compliance OCC 5: core winding variation DET 4: scale test detects most cases
Sew Cups	Uneven cup alignment pre-sew	8	4	5	160	MODERATE	SEV 8: affects seam symmetry OCC 4: experienced sewers reduce risk DET 5: visual check before sewing

ACTION PLAN

Recommendations

Priority	Action	Target
IMMEDIATE RPN 336	Stitch Counter + Alignment Jig — install stitch counter tool; introduce cup-alignment jig for pre-sew positioning. First-article inspection SOP mandatory per batch.	Stitch DPU → <0.02
IMMEDIATE RPN 324	Calibrated Bounce/Size Testing — replace founder visual/feel inspection with calibrated bounce height fixture and go/no-go size gauge. SPC control charts on bounce height.	Rejection rate → <2%
SHORT-TERM Step 3 DPU	Print Logo Standardisation — Step 3 DPU 0.16 vs avg 0.08. Introduce logo placement jig, standardise heat seal settings (temperature, dwell time, pressure). Error-proof station.	Step 3 DPU 0.16 → <0.05
SHORT-TERM RPN 192	Incoming Material Inspection — no controls on supplier leather. Incoming goods checklist, supplier scorecard, approved supplier list with quarterly review.	0 substandard batches
ONGOING	RTY Improvement Programme — if Step 3 DPU → 0.04 and stitch defects halved, RTY improves from 66% to ~88%. Recalculate RTY quarterly.	RTY: 66% → 88% in 6mo
ONGOING	Quality Dashboard — FTY and DPU per step tracked weekly. I-MR / p-charts for weight and bounce. CAPA trigger if any step FTY drops below 90%.	Dashboard live in 3 wks

BUSINESS VALUE

Financial Impact & Expected Benefits

66%→88%	0.16→<0.05	RPN 336→<80	528→~230
RTY Target	Step 3 DPU	Post-FMEA Target	Projected Defects

Improvement	Current	Target	Business Impact
RTY improvement	66%	88%+	34% fewer defective units — lower scrap, rework, and overtime
Step 3 (Print Logo)	DPU 0.160	DPU <0.05	256 → ~80 defects/1,600 units = 176 fewer reworks per batch
Stitch failure (RPN 336)	No stitch counter	Stitch counter + jig	Eliminates highest-risk mode; prevents seam separation in play
Bounce/size testing	Visual/feel only	Calibrated fixture + gauge	Objective, repeatable test — eliminates operator-dependent variation
Supplier material	No incoming inspection	Incoming QC protocol	Prevents substandard batches; reduces wasted downstream processing

- **Improving RTY from 66% to 88%** means 22 additional defect-free units per 100 produced — reducing rework labour, scrap material cost, and overtime to meet production targets.
- **Step 3 standardisation is the highest-ROI action:** it contributes the most to RTY drag and requires only tooling and procedure changes — no capital investment required.
- **Calibrated testing at the Bounce/Size step** converts a subjective, founder-dependent decision into an objective, SPC-capable measurement — creating long-term, auditable process control.

REFLECTION

Key Insights & Future Opportunities

Key Insights

- **C&E Matrix and FTY/DPU told different stories — both were needed:** The C&E Matrix flagged Bounce/Size Test and stitching as the highest-impact design inputs. FTY/DPU independently revealed Print Logo (Step 3) as the worst-performing step operationally. Neither tool alone gives the full picture.
- **FMEA RPN drives prioritisation, not gut feel:** RPN ranking (336 → 324 → 210 → 192 → 180 → 160) gave a defensible, data-driven corrective action sequence across 6 failure modes — removing subjectivity from improvement decisions.
- **RTY is more honest than step-level FTY:** Each step appeared capable (84–96% FTY). Multiplying them together to get RTY = 66% showed that process quality compounds — small inefficiencies create a large aggregate gap.
- **Poka-yoke over training:** RPN 336 was driven by manual counting error and fatigue. A stitch counter tool makes the error physically impossible — a permanent fix, not a temporary behaviour change.

Future Opportunities

- Re-run the C&E Matrix and FMEA post-improvement to recalculate RPN values and confirm risk reduction — closing the PDCA loop.
- Implement SPC on bounce height (I-MR chart) to convert the current pass/fail attribute inspection into a variable measurement with early drift detection.

Core Competencies Applied

Skill / Competency	Evidence in This Project
SIPOC	Full 8-stage SIPOC map; established scope boundary for C&E Matrix scoring
C&E; Matrix (full grid)	7 requirements weighted (7–10); 8 steps scored 0/1/3/9; vital few identified by ranked totals
FMEA — SEV/OCC/DET Rationale	SEV/OCC/DET rated per failure mode with rationale; RPN calculated; actions ranked by priority
FTY & DPU	FTY & DPU calculated from raw defect data per step; Step 3 outlier clearly identified
RTY (Rolled Throughput Yield)	RTY = 66% = product of 5 step FTYs; each step contribution to yield loss quantified
Pareto / Vital Few	C&E totals and FMEA RPNs ranked; 80/20 applied to focus improvement effort